MILITARY SPECIFICATION

SLEEVING, FOR IDENTIFICATION MARKING, HEAT SHRINKABLE, GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification establishes the requirements for various types of identification sleeving that will shrink to a predetermined size upon the application of heat after it has been marked. Continuous operating temperature ratings range from -55° to 175°C (-67° to 347°F). (See 6.1).

1.2 Part numbers. Part numbers under this specification are coded as follows:

M85848/XX - X X XX -X

- XX: Applicable specification sheet identifier
- X: Type (See applicable specification sheet)
- X: Class
- XX: Size identifier
- -X: Color designator (See 1.3)

See applicable specification sheet for example.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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1.3 Color. The color shall be identified in accordance with the following:

Designator	Color	Designator	Color
0	Black	6	Blue
1	Brown	7	Violet (Purple)
2	Red	8	Gray (Slate)
3	Orange	9	White
4	Yellow	C	Clear
5	Green	P	Pink

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-S-1926 - Sodium Chloride, Technical
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Box, Fiberboard, Corrugated, Triple Wall
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8243 - Anti-Icing and Deicing-Defrosting Fluid
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
MIL-M-81531 - Marking of Electrical Insulating Materials

(See Supplement 1 for applicable specification sheets).

STANDARDS

FEDERAL

FED-STD-228 - Cable and Wire, Insulated, Methods of Testing

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STANDARDS (Continued)

MILITARY

MIL-STD-104 - Limit for Electrical Insulation Color
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 570 - Testing for Water Absorption of Plastics
ASTM D 638 - Tensile Properties of Plastics
ASTM D 792 - Specific Gravity and Density of Plastics by Displacement
ASTM D 882 - Tensile Properties of Thin Plastic Sheeting
ASTM D 2671 - Testing Heat Shrinkable Tubing for Electrical Use
ASTM D 3951 - Commercial Packaging
ASTM E 595 - Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment
ASTM G 21 - Resistance of Synthetic Polymeric Materials to Fungi, Recommended Practice for Determining

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Specification sheets. The heat shrinkable identification sleeving shall be as specified herein and in accordance with the applicable specification sheets. In the event of discrepancy between this specification and the specification sheet, the specification sheet shall govern.

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3.2 First article. When specified in the contract or purchase order, a sample will be subjected to first article inspection (See 6.2.2).

3.3 Dimensions and Tolerances. See specification sheets for dimensional and tolerance requirements.

3.3.1 Type I-Flatweb format.

3.3.1.1 Sleeve length and width "as supplied". When tested as specified herein, the sleeve length and width shall meet the requirement of the applicable specification sheet.

3.3.2 Type II-Sleeving. When tested as specified herein, the as supplied inside diameter, after recovery inside diameter, and the length shall meet the requirement in the applicable specification sheet.

3.3.3 Unit of issue. The unit of issue for identification sleeving shall be an individual piece. Reference shall be made to the applicable specification sheet for the minimum quantity of sleeving pieces available from a package.

3.3.4 Sheets and rolls. Unless otherwise specified in the contract or order, the minimum quantity of sleeving available in standard sheet or roll form shall be as specified in the applicable specification sheet.

3.4 Color. The "as supplied" color of the sleeving shall be in accordance with the requirements of MIL-STD-104. MIL-STD-104 does not include any limits for pink or tan colors. Munsell color notations for these colors shall be as follows:

Color	Light limit	Nominal	Dark limit
Pink	3.75R 6.5/9.0	2.5R 6.0/7.0	1.25R 5.5/6.0
Tan	5Y 5.5/6.0	5Y 5.0/5.0	5YR 4.5/4.0

3.5 Restricted shrinkage. After testing as specified in 4.6.3, the shrunk sleeving or flat web material shall fit snugly over the wire without cracking.

3.6 Properties.

3.6.1 Color stability. When tested as specified in 4.6.4, the sleeving shall be within the "as received" limits for the color in MIL-STD-104.

3.6.2 Tensile Strength. Tensile strength shall be as specified in the applicable specification sheet when determined in accordance with 4.6.5.

3.6.3 Modulus. See specification sheet for secant modulus or tensile stress as applicable. Testing shall be as specified in 4.6.6 and may be performed on as supplied or shrunk sleeves.

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3.6.4 Heat shock. When tested as specified in 4.6.7, the sleeving shall show no sign of cracking, flowing, or dripping. After bending as specified in the applicable specification sheet, the sleeve shall show no sign of cracking except that cracking at the point of bend shall not be cause for rejection. Any cracking sufficient to cause a loss of any part of a marker legend shall be cause for rejection.

3.6.5 Heat resistance. After testing as specified in 4.6.8, the specimens shall pass the print performance evaluation specified therein and the bend test requirements in 3.6.4 and the applicable specification sheet.

3.6.6 Low temperature flexibility. When tested as specified in 4.6.9, the sleeving shall meet the requirement of the applicable specification sheet.

3.6.7 Fluid resistance. When tested as specified in 4.6.10, the sleeving shall meet the requirement of the applicable specification sheet

3.6.8 Fungus Resistance. Fungus resistance ratings shall be as specified in the applicable specification sheet when tested as specified in 4.6.1.

3.6.9 Specific gravity. Specific gravity determinations shall be in accordance with the applicable specification sheet when determined as specified in 4.6.1.

3.6.10 Water absorption. Water absorption shall be as specified in the applicable specification sheet, when tested in accordance with 4.6.1.

3.6.11 Flammability. The marking material shall pass the applicable flammability test of 4.6.11 specified in the specification sheet.

3.7 Vacuum outgassing. When tested as specified in 4.6.12, the sleeving shall meet the requirement of the applicable specification sheet.

3.8 Print performance. When tested as specified in 4.6.13, the sleeving shall meet the requirement of the applicable specification sheet.

3.9 Storage requirements

3.9.1 Shelf life. The manufacturer shall certify in writing that the sleeving, after storage at the conditions specified in the applicable specification sheet, shall conform to the "as supplied" sleeve width or diameter and length requirements of the applicable specification sheet.

3.9.2 Extension of shelf life. The expiration date may be extended by a period of time equal to 50% of the original shelf life for the sleeving if the dimensions are still within specification limits when tested as specified in 4.6.14.

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3.10 Workmanship. When examined visually, the sleeving, both before and after shrinkage, shall be free from internal voids, blisters, tears, seams, cracks, foreign matter or other defects that would be detrimental to fabrication, appearance, or performance. After shrinkage, the sleeving surface shall be smooth and uniform. When the sleeving contains a single or double bonded seam, the overlaps and seams normally associated with these manufacturing techniques shall be accepted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The examination and testing of the sleeving shall be classified as follows:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

4.3 First article inspection. First Article inspection shall consist of all the tests in Table I and inspections of this specification. The First Article inspection shall be performed on the first lot or order of identification sleeving furnished under this specification and on any subsequent lot or order when specified by the acquisition activity (see 4.3.1 and 6.2). Instructions for testing and approving the sleeving are located in 6.2.2.

4.3.1 Prior approval. If a contractor has previously delivered an acceptable product meeting the requirements of this specification, First Article inspection may be waived at the discretion of the acquiring activity for a period of time not to exceed 2 years.

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4.3.2 First Article sample. The supplier shall offer a sufficient quantity of material to complete the entire First Article test spectrum. The shrinkable sleeving shall be of the type, class, and size for which approval is desired . Approval of the sample will automatically extend First Article approval to all sizes in the range specified in Table II. The First Article sample shall be produced using the same plant facilities intended for the production run.

4.3.2.1 First article sample identification. The First Article sample shall be plainly identified by securely attached durable tags or labels marked with the following information:

Sample for First Article Inspection
MIL-S-85848(AS) (Add Specification Sheet Number)
SLEEVING, FOR IDENTIFICATION MARKING, HEAT SHRINKABLE
Name of Manufacturer
Product Code Number
Date of Manufacture
Submitted by (name)(date) for First Article Inspection in accordance with the requirements of (give specification sheet number, class, and part number as applicable)

4.3.3 Manufacturer's data. For each acquisition, the manufacturer shall submit two copies of his test report that show the sleeving conforms to the requirements of the specification. The data shall include certification that the sleeving meets or exceeds the shelf life established in the applicable specification sheet.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Unless otherwise specified, a lot shall consist of all the heat shrinkable identification sleeving of one specification sheet, class, color, size and length that has been processed under essentially the same conditions into a finished form from one batch and available for inspection at one time. A batch shall be the quantity of material mixed, blended, or processed in a single operation intended to make the product uniform. If the material cannot be identified by batch, a lot shall consist of sleeving from one specification sheet, class, color, size and length that has been produced at one plant under essentially the same conditions, processed in a continuous operation and available for inspection at one time.

4.4.2 Sampling.

4.4.2.1 Visual and dimensional inspection of end item. Samples for this inspection shall be selected in accordance with Inspection Level S-3 of MIL-STD-105. Unless otherwise specified in Table III, the sampling unit shall be 1 foot of sleeving or 1 linear foot of flatweb format sleeving. Each sample shall be inspected as specified in 4.4.3.1.

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a. One unit of product for the examination specified in

4.4.2.2 Physical property tests. Sufficient heat shrinkable material shall be selected at random from each lot (4.4.1) for the tests specified in 4.4.3.2. Sleeving selected and visually inspected in 4.4.3.1 may be used to "make-up" the sample size for the tests.

4.4.2.3 Packaging. A quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random from each lot in accordance with Inspection Level S-2 of MIL-STD-105. The lot size shall be the number of shipping containers.

4.4.3 Inspection and tests.

4.4.3.1 Visual and dimensional inspections. Samples selected in 4.4.2.1 shall be examined for conformance to the requirements in Table III. The Acceptable Quality Level (AQL) for this inspection shall be as specified in Table III. The number of determinations of each sample shall be as specified in Table I.

4.4.3.2 Physical property tests. Test specimens prepared from the sample selected in 4.4.2.2 shall be tested to the requirements of Table IV. The number of specimens tested and reporting of results shall be in accordance with Table I. Nonconformance of the sample to a single requirement shall be cause for rejection of the lot represented by the sample.

4.4.3.3 Packaging inspection. Each sample selected in accordance with 4.4.2.3 shall be examined for the defects specified in Table V and to all other requirements of Section 5 of this specification. The AQL for this inspection shall be 4.0. In addition, shipping containers fully prepared for delivery shall be examined after closing for closure defects.

4.4.4 Reporting of test results. Tables III and IV shall be used to prepare test reports. The number of specimens per test and method of reporting the results are contained in Table I.

4.5 Test and unrestricted shrinkage procedures.

4.5.1 Test conditions. Unless otherwise specified herein, the heat shrinkable sleeving and measurement gages shall be conditioned for 4 hours at 23°+ 1°C (73.4°+ 2°F) and a relative humidity of 50 ± 5 percent prior to testing. When the sleeving is tested after heat shrinking, conditioning of the sleeving shall take place after completion of the heat shrinking procedure. Unless otherwise specified in the specification sheet, an oven with a circulating air velocity of 100 to 200 feet per minute or a 1,000 watt, 600° to 700°F air flow temperature heat gun shall be used whenever heating is required.

4.5.2 Unrestricted shrinkage. When unrestricted shrinkage is required for a test method, it shall be accomplished by one of the following methods:

4.5.2.1 Unsupported method. Lay an appropriate length of identification sleeving in a tray of an oven at the temperature and for the time specified in the specification sheet. If the sleeving becomes tacky in the oven, a small amount of powdered talc may be placed on the tray to prevent sticking.

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4.5.2.2 Mandrel method. Select a smooth, clean, metallic mandrel that has a diameter equal to the maximum acceptable sleeving I.D. (after unrestricted shrinkage) minus 0, plus 0.002 inch, or 2 percent, whichever is less. Slip an appropriate length of sleeving on the mandrel and heat in an oven at the temperature and for the time specified in the applicable specification sheet.

4.6 Test methods.

4.6.1 ASTM tests. ASTM tests shall be performed in accordance with ASTM Standard Methods of Test (see 2.2) with any exceptions that may be specified herein or in the specification sheets.

4.6.2 Dimensions. The "as supplied" length and width shall be measured in accordance with ASTM D 2671.

4.6.3 Restricted shrinkage. A PTFE insulated wire having a diameter that is below the midpoint of the range recommended in the applicable specification sheet for the size being tested shall be prepared for each sleeving. The length of the sleeving shall be placed over the wire and shrunk in a forced convection oven. The heating time in the oven shall be as specified in the applicable specification sheet. Upon removal from the oven, the specimens shall be conditioned as specified in 4.5.1, then visually examined for cracking and fit. Any cracking of the identification sleeving or movement away from the wire shall be cause for rejection.

4.6.4 Color stability. An appropriate length of identification marking material shall be shrunk as specified in 4.5.2, then placed in an air circulating oven for the time and for the temperature specified in the applicable specification sheet. After removal from the oven, the specimens shall be cooled, conditioned as specified in 4.5.1. and examined for conformance to the color requirement of the applicable specification sheet.

4.6.5 Tensile strength. Tensile strength properties shall be determined in accordance with ASTM D 638 at the speeds specified in the applicable specification sheet. The marking sleeving, or a 1 by 6 inch piece of material representive of the sleeving being tested shall be shrunk as specified in 4.5.2. One inch bench marks and a one inch jaw separation shall be used. Requirement shall be as specified in the specification sheet.

4.6.6 Modulus.

4.6.6.1 Secant modulus. Full sections of sleeving or a 1 by 6 inch piece of material representative of the sample shall be tested using the procedure and the requirement as specified in the specification sheet.

4.6.6.2 Tensile stress. Full sections of sleeving or a 1 by 6 inch length of material representative of the sample being tested shall be shrunk as specified in 4.5.2. Test procedure, along with percent strain, and the requirement shall be as specified in the specification sheet. One inch bench marks and 2 inch initial jaw separation shall be used. A break at the bench mark or outside the gage length shall be cause for retest.

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4.6.7 Heat shock. Three test specimens shall be prepared, first by printing as specified in 4.6.13, then by shrinking the marked material onto a 10 inch length of PTFE insulated wire as specified in 4.6.3. The insulated wire shall be of a diameter that is within the range recommended for the size of marking material. The specimens shall be suspended vertically, in an air circulating oven for a period of four(4) hours at the temperature in the applicable specification sheet. While in the oven and after removal from the oven, the specimen shall be examined for evidence of cracking, flowing, or dripping. The specimen shall be conditioned as specified in 4.5.2 followed by a bend test. The specimen shall be grasped within 2 inches of each side of the material and be subjected to a bend/straightening cycle through the number of degrees specified in the applicable specification sheet in 10 ± 2 seconds. The sleeve shall be positioned at the center of the bend. Cracking at the point of the bend shall be permitted, however there shall be no separation of the sleeve at the seam. The sleeve shall then be evaluated for conformance to 3.6.4.

4.6.8 Heat resistance. Three 2 inch lengths of marking material shall be printed using the manufacturer's recommended practice. Each marked material shall be shrunk onto a 10 inch length of PTFE insulated wire as specified in 4.6.3. The insulated wire shall be of a diameter that is below the midpoint range/diameter of the sleeve size being tested. The specimens shall be placed in an oven for the time and temperature specified in the applicable specification sheet. After cooling to room temperature, the specimens shall be evaluated for print performance in accordance with MIL-M-81531(para.4.6.2), followed by the bend test specified in 4.6.7 herein. Conformance to 3.6.5 and the specification sheet shall be noted.

4.6.9 Low temperature flexibility. Test specimens shall be prepared as specified in 4.6.7. The specimens shall be conditioned in a cold chamber for 4 hours at the temperature specified in the applicable specification sheet. After completion of the conditioning period and while still in the cold chamber, the specimen shall be grasped at each side within 2 inches of the marking material and bent/straightened through the number of degrees specified in the specification sheet within 10 ± 2 seconds. Visually examine the material for cracks. Cracking at the point of bend shall be permitted, however there shall be no separation along any seam. Any cracking sufficient to cause a loss of any part of the marking legend shall be cause for rejection.

4.6.10 Fluid resistance. The identification marking shall be shrunk and conditioned as specified in 4.5. A suitable number of required lengths of sleeving or 1 by 6 inch lengths of material representative of the sleeve being tested for tensile strength and at least three 2-inch lengths of sleeves printed in accordance with the manufacturer's recommended practice shall be totally immersed in each of the six fluids specified in Table VI. The volume of the fluid shall not be less than 20 times that of the sleeving. After 24 hours immersion at 24°+ 3°C (75°+ 5.4°F), the sleeving shall be lightly wiped and then conditioned for 45 ± 15 minutes at room temperature. The sleeving from each of the test fluids shall then be tested for conformance to the tensile strength requirement of the specification sheet. The printed specimens shall be evaluated as specified in MIL-M-81531(para. 4.6.2) for conformance to the specification sheet requirement.

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4.6.11 Flammability. One of the following tests shall be performed as specified in the specification sheet. Sleeving that passes test C has better flame resistant properties than sleeving that passes either test A or B; and sleeving that passes test B has better flame resistant properties than sleeving that passes test A.

4.6.11.1 Test A. An 18 inch (0.46 m) specimen of marking material shall be shrunk as specified in 4.6.3, sample preparation A or B, onto a 21 inch (0.53m) metal rod. The outside diameter of the metal rod shall be the maximum recovered inside diameter of the sleeving specified in the applicable specification sheet. the specimen so prepared shall be tested in accordance with FED-STD-228 method 5221 for vertical flammability. Brown scorching or soot on the indicator flag shall not be cause for rejection.

4.6.11.2 Test B. A 22 inch (0.55 m) specimen of marking material shall be shrunk as specified in 4.5.2 and tested in accordance with ASTM D 876.

4.6.11.3 Test C. A 26 inch (0.66 m) length of marking material shall be shrunk on a 0.015 to 0.032 inch (0.38 by 0.81 mm) spring steel music wire. For flatweb sleeves, the sleeves may be nested to simulate a 26 inch continuous length. There shall be an air gap between the wire and shrunk down sleeving. The specimen shall then be mounted in the same manner and test equipment described in 4.6.10.1. The sleeving shall be mounted by clamping the top so there shall be no chimney effect. There shall also be a flat horizontal layer of untreated surgical cotton placed on the floor on the enclosure (entered under the specimen 9-1/2 inches (24.1 cm) below the point of flame application). The specimen so prepared shall then be tested in accordance with FED-STD-228 Method 5221 for vertical flammability, with the following exception:

4.6.11.3.1 Test C Procedure. The valve supplying gas to the burner is to be opened to apply the flame to the specimen automatically. This valve is to be held open for 15 seconds, closed for as little as 15 seconds, opened for 15 seconds, etc. for a total of five 15 second applications of the gas flame to the specimen with as little as 15 seconds between applications. In no case is the gas flame to be reapplied to the specimen until any flaming or glowing of the specimen from the previous application of the gas flame ceases of its own accord even though waiting for the specimen to cease flaming or glowing may result in more than 15 seconds between applications of the gas flame. The duration of flaming or glowing of the specimen after each removal of the gas flame is to be recorded.

4.6.11.3.2 Test C Observation. Note shall be made and recorded of whether any flaming or glowing particles or flaming drops fall from the specimen. If more than 25 percent of the indicator flag is burned away or charred (brown scorching and soot shall be ignored) after all of the five applications of flame or if flaming or glowing particles or flaming drops fall from the specimen at any time and ignite the cotton (flameless charring of the cotton is to be ignored), the specimen shall be judged to have conveyed flame. Any specimen which continues to flame or glow from one flame application longer than one minute shall not be acceptable.

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4.6.12 Vacuum outgassing (as supplied). Testing for outgassing shall be conducted in accordance with ASTM E 595. Results shall comply with the requirements of the applicable specification sheet.

4.6.13 Print performance. Twenty-four 2 inch lengths of marking material shall be printed using the supplier's recommended practice. Twelve specimens shall be shrunk (4.5.2) onto each of twelve wires having diameters that fall within the recommended ranges in the applicable specification sheet. The remaining twelve printed specimens shall be fitted, unshrunk, onto individual insulated wires. Each specimen shall be snug on the wire. All specimens shall be conditioned as specified in 4.5.1. for 24 hours. Three specimens from the shrunk, and three from the unshrunk specimens shall be evaluated in accordance with paragraph 4.6.2 of MIL-M-81531. The remaining specimens shall be evaluated as specified in Method 215 of MIL-STD-202 except that solvent solution "c" in that method shall not be used. Conformance to the applicable requirement in the specification sheet shall be noted.

4.6.13 Extension of Shelf Life Expiration Date. The following examination shall be performed to determine if an expired shelf life date may be extended. The as-supplied width and length shall be measured in accordance with ASTM D 2671. The measured dimensions shall be compared to the requirements given in the appropriate construction table of the specification sheet.

5. PACKAGING

5.1 Preservation-packing. Preservation-packaging shall be Level A or Industrial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Heat shrinkable identification sleeving shall be packaged Method III in accordance with MIL-P-116 and shall be furnished in straight lengths or sheet form, on coils, on spools, on spool or reel type packages, or on center dispensing coil type packages. Only identical sleeving (type, class, size, color, etc.) shall be contained in any one coil, spool, spool or reel type package, or center dispensing coil type package. Unless otherwise specified (see 6.2.1), there shall be no more than 3 lengths of sleeving on a spool, in a spool or reel type package, or in a center dispensing coil type package; and the length of any piece shall be not less than 25 percent of the total footage on each spool or in each package. Unless otherwise specified (see 6.2.1), a quantity of sleeving considerably less than the manufacturer's standard quantity normally wound on a spool, in a spool or reel type package, or in a center dispensing coil type package may be supplied in a coil of one continuous length.

5.1.1.1.1 Straight lengths or sheet form. Straight lengths or sheet form sleeves shall be bundled and shall be firmly tied or taped at each end.

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5.1.1.1.2 Coils. Coils of heat shrinkable identification sleeving shall be uniform, compact, and of sufficient diameter to prevent creasing or kinking of the sleeving. Each individual coil shall be firmly tied, taped, or otherwise secured at a minimum of three places equidistantly spaced around the circumference of the coil.

5.1.1.1.3 Spools. Unless otherwise specified (see 6.2), spools shall be of a type, size, construction, and material normally used for the product in good industry practice. The heat shrinkable identification sleeving shall be evenly wound on spools in accordance with good industry practice to obtain uniformity, compactness, and nesting of successive turns and layers. The end of the sleeving shall be secured by a suitable means to prevent unwinding.

5.1.1.1.4 Spool or reel type packages. Spool or reel type packages for containing and dispensing heat shrinkable identification sleeving shall be of fiberboard, plastic, or composition material and shall be completely self-contained protective units or boxes from which the sleeving can be withdrawn without disassembly of the unit or box. The construction of the package shall be such as to provide either a stationary or movable spool or reel within the unit or box. The end of the heat shrinkable identification sleeving shall be secured to prevent unwinding.

5.1.1.1.5 Center dispensing coil type package. Center dispensing coil type package shall be of the type, size, construction, and material normally used for the product in good industry practice. Generally, this package is a corrugated fiberboard box with a perforated disc in one side. When dispensing is desired, the disc is removed and the end of the sleeving is withdrawn through the hole. Continuous dispensing is made from the inside circumference of the coil of sleeving packaged in the box.

5.1.2 Industrial. Heat shrinkable marking material shall be packaged in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, or Industrial, as specified (see 6.2.1). Exterior containers shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practical.

5.2.1 Level A. Heat shrinkable marking material packaged as described in 5.1 shall be packed in fiberboard containers conforming to:

a. PPP-B-636, class weather resistant, style RSC, special requirement. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with an exterior waterproof tape, 2 inches (5.1 cm) minimum width, conforming to PPP-T-60. Banding (reinforcing requirements) shall be applied in accordance with the appendix to PPP-B-636 using tape banding only.

b. PPP-B-640, Class 2, style RSC, closure and reinforcing shall be in accordance with the appendix therein. When more than one coil or spool is packed to a shipping container, the coils or spools shall be separated both vertically and horizontally by fiberboard sheets.

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5.2.2 Level B.

5.2.2.1 Straight lengths, coils, spools. Straight lengths, coils, and spools of heat shrinkable marking material shall be packed in containers conforming to PPP-B-636, class-domestic, style RSC, special requirement or PPP-B-640, Class 1. Closure and reinforcing shall be in accordance with the applicable container specification.

5.2.2.2 Spool or reel type package or center dispensing coil type package. Packages as described in 5.1.1.1.4 and 5.1.1.1.5 generally do not require additional packing for shipment since the unit package may serve also as the shipping container; however, to facilitate handling, loading, etc. small packages may be packed as described in 5.2.2.1.

5.2.3 Industrial. Heat shrinkable marking material shall be packed in accordance with ASTM D 3951.

5.3 Unitized loads. Unitized loads should be used when practical and shall be commensurate with the level of packing stated in the contract or order (see 6.2.1). MIL-STD-147 shall be used when loads are palletized.

5.4 Marking. Coils, packages, shipping containers, and unitized loads shall be marked in accordance with MIL-STD-129.

5.4.1 Additional markings.

a. Date manufactured.
b. Expiration date.
c. Lot, batch or control number.
d. Special marking as required in the specification sheet.
e. Other markings required by the contract or order.

6. NOTES

6.1 Intended use. Heat shrinkable identification marking sleeving is intended for use as a snug-fitting identifier over insulated wires, insulated wire bundles, cables and cable harnesses, metal or fibrous tubing, connectors and terminals. Specific property values in Section 3 and in the specification sheets should be evaluated for the particular use intended. Heat shrinkable identification marking sleeving should be marked prior to shrinking and be used to identify materials that cannot be marked directly. The sleeving shall not act as a primary insulator. See the specification sheet for specific end uses.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification, applicable specification sheet, sleeving type and class.
b. Part number (including color designation) (see 3.2.2).

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c. Quantity, total number of pieces for identification sleeving (unit of issue = pieces).
d. Total number of pieces on sheet or roll form for identification sleeving.
e. Specific gravity if required (see 3.5.2).
f. Tests specified by procuring activity (see applicable specification sheet).
g. Waiver or requirement for number of individual lengths on a spool is required for a specific type of sleeving (see 5.1.1.3).
h. Levels of packaging and packing (see 5.1 and 5.2).
i. Type, size, construction, and material or spool if other than specified (see 5.1.1.3.2).
j. First Article approval, if required (see 3.3 and 4.3).

6.2.2 Contracts or orders shall specify the following provisions for First Article inspection (see 4.3.1):

6.2.2.1 Whether First Article inspection is required. When a contractor is in continuous production of the sleeving from contract to contract, consideration should be given to waive the First Article inspections within the time limitations specified in 4.3.1. If inspection is required, indicate:

a. Where the First Article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

b. Scheduling of First Article inspection. As soon as practicable during the course of the contract, the contractor shall prepare for First Article testing. Prior to the start of testing, the contractor shall notify the procuring activity in sufficient time to allow representation during the tests.

c. That the approval of First Article samples or the waiving of the First Article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Subject term (keyword) listing.

First article
Heat shrinkable
Identification marking
Non-insulating
sleeves

Preparing Activity
Navy- AS
(Project No. 5970-N664)

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TABLE I. First Article Inspection

Property	Requirement	Number of determinations per sample	Results Reported As	
			Pass or Fail 1/	Average of Multiple Determinations 2/
Dimensions	3.3	Three 3/	X	
Color	3.4	One	X	
Restricted shrinkage	3.5	Two	X	
Color stability	3.6.1	Three	X	
Tensile strength	3.6.2	Five		X
Modulus	3.6.3	Five		X
Heat shock	3.6.4	Three	X	
Heat resistance	3.6.5	Three	X	
Low temperature Flexibility	3.6.6	Three	X	
Fluid resistance	3.6.7	Five		X
Fungus resistance	3.6.8	Three	X	
Specific gravity	3.6.9	Three		X
Water absorption	3.6.10	Three		X
Flammability	3.6.11	Three	X	
Vacuum outgassing	3.7	One	X	
Print performance	3.8	One	X	
Shelf life	3.9.1	Three	X	
Shelf life extension	3.9.2	Three	X	
Workmanship	3.10	Three	X	

1/ Nonconformance of one determination shall constitute failure of the sample. A description or the numerical point of the failure shall be recorded also.
2/ All values on which the average is based shall be recorded.
3/ Three specimens for each dimension.

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TABLE II. First Article Test Groups

Group	Size to fit wire diameter
1	Up to 0.159 inch
2	0.160 to 0.375 inch
3	0.376 inch and above

TABLE III. Visual and Dimensional Inspection

Inspection	Requirement	AQL
Workmanship As Supplied	3.10	2.5
Spool, Coil Length 1/ 2/	3.3.4	
Sleeve Width, or diameter	3.3.1	1.0
Sleeve Length	3.3.1	1.0
After Unrestricted Shrinkage Color	3.4	Zero Defect

1/ The lot size shall be the number of spools, coils, etc.; the sample unit-one spool.

2/ If the average length per spool of the selected samples is less than the quantity per spool specified in the contract or order, the lot represented by the sample shall be rejected.

TABLE IV. Physical Property Tests 1/

PROPERTY	TEST PARAGRAPH
Tensile strength	4.6.5
Heat Shock	4.6.7
Low temperature flexibility	4.6.9
Flammability	4.6.11

1/ The acquiring activity, at its option, may require conformance to any additional requirements specified.

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TABLE V. Packaging Inspection.

EXAMINE	DEFECT
Packaging	Not level required by contract or purchase order. Material or construction not as specified.
Packing	Not level required by contract or purchase order. Material or construction not as specified. Any nonconforming component, incomplete closures, inadequate strapping, bulged or damaged containers.
Count	Less than specified or indicated quantity per shipping container.
Markings	Warning labels or marking missing. Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

TABLE VI. Test Fluids.

FLUID	CONFORMING TO
Hydraulic fluid, petroleum base	MIL-H-5606
JP-4 Fuel	MIL-T-5624
Lubricating Oil	MIL-L-7808
Lubricating Oil	MIL-L-23699
5 Percent NaCl	O-S-1926
Deicing Fluid	MIL-A-8243

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NAVY

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-55543 (NAVY)	SLEEVING, FOR IDENTIFICATION MARKING, HEAT SHRINKABLE, GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
5. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

6. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

7. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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